# Visual Kids: Interactive Learning Application for Hearing-Impaired Primary School Kids in Sri Lanka


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***Abstract*—E-learning is one of the most common methods used for educational purposes. However, most e-learning environments are not particularly useful for students with hearing impairments due to their lack of adaptability. "Visual Kids" is an e-learning application that provides deaf students with an accessible learning experience through video sign language tutorials, visual aids, and interactive activities. According to the primary educational curriculum, this platform will show students how to write letters and numbers step-by-step. Also, this application includes a gamified learning environment with interactive games and quizzes that make learning fun and enjoyable. This application provides the capability of teaching letters and numbers using sign language, and it recognizes the sign language, handwritten letters and numbers that students provide. While answering the game-based questions, the student's facial expressions can change to happy, sad, or angry. Their expressions can be identified using face recognition technology. Screening results from the models built in this research provided an accuracy of 96%, 98%, and 98% for sign language recognition, identifying handwritten letters and numbers, and identifying emotions using facial expressions, respectively. This application aims to improve educational outcomes for deaf primary pupils and support their academic and social progress.**



***Index Terms*—Hearing Impaired students, Sign language, E-learning application, Face Emotion Recognition, Game-Based Learning, Handwritten Recognition**


## I. Introduction

Hearing is the ability to perceive and interpret sound waves in the environment. Hearing allows us to sense and understand the auditory stimuli around us, which plays a crucial role in communication, learning, and our overall perception of the world. However, due to hearing problems, some groups of people are unable to listen. This will eventually make it difficult for them to share their ideas with others and get the education they need.

Hearing impairment (HI) or hearing loss is an inability to hear. Hearing loss may be mild, moderate, severe, or profound [1]. The condition affects one or both ears and causes difficulty listening to conversational speech or loud noises. Hearing loss can be due to different causes, such as childhood or infant hearing loss, a variety of infections, such as chronic ear infections, certain toxins or medications, noise-induced hearing loss, and ageing-related hearing loss. Depending on the severity of the hearing loss, it can be temporary or permanent [1]. As a result, this deafness can have a significant impact on an individual's daily life, affecting their ability to communicate and participate in activities.

In recent years, hearing loss has become a major issue in many nations. The World Health Organization (WHO) estimates that over 466 million individuals worldwide have hearing issues, per the article [1]. 432 million of these 466 million people are adults, and 34 million are kids. Similar to other countries in the world, Sri Lanka also suffers from a hearing disability. According to the article, the WHO estimates that 9% of Sri Lankans have hearing problems [2]. The main problem that should be considered is that many children are part of this deaf community, while there are school-going children among them. In Sri Lanka, 4% of primary school students have hearing loss disabilities [3]. These children are exposed to different barriers and challenges when obtaining an education. Societal attitudes, a lack of understanding, a lack of access to tools for education and essential services, a shortage of specialized teaching and learning resources and assistive devices, limited access to training tools, and communication


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issues are some of the challenges these students face in learning [4]. In general, these barriers arise from not being able to communicate with each other and not being capable of learning very quickly like other children.

The usage of technology within the education department has become more vital with each passing year. The E-Learning environment is one of the most popular technical approaches used. In education, e-learning environments are used by a wide variety of students, including those who are hearing impaired. However, the Sri Lankan deaf community has access to only limited applications and research designed specifically for them. The previous applications for hearing-impaired people were only to help them recognize sign language [11,12]. Those applications could not help children learn various kinds of subjects, as in public schools. Without an online tutoring application for children with hearing impairments, these children will increasingly lag behind normal children.

This research focuses on studying and developing an e-learning application for primary students with hearing impairments, specifically targeting children in the age range of 5 to 10 years old. This research aims to help these students learn basic letters and numbers using sign language, as well as to facilitate easier letter and number writing. The research seeks to create an e-learning application designed for the convenience of hearing-impaired primary students, allowing them to independently enhance their knowledge without the need for constant guidance from teachers or parents. For this research, chose hearing-impaired primary students because they are in a crucial stage of language development, making early sign language intervention beneficial for their language skills and overall well-being. Primary school students with hearing impairments may also be more motivated to learn than older students, as they are still developing their language skills and may be eager to learn a new way to communicate.

## II. Literature Review

The advancement of technology has a better way for innovative educational approaches, including e-learning, which has the potential to enhance the learning experiences of students with diverse needs [5]. In the case of hearing-impaired children, e-learning applications can offer valuable support in their educational journey. Some authors have worked in the past to provide solutions to avoid learning difficulties by creating various e-learning applications using different technologies to teach hearing-impaired children.

In the hearing-impaired kid's project, one of the objectives is teaching English letters and numbers using sign language [6]. There are several previous studies that have addressed the importance of teaching sign language to hearing-impaired children. There is research on e-learning environments specifically designed for hearing-impaired students, and the study emphasized the significance of incorporating sign language into the curriculum to facilitate communication and language development [6].

Another study, which demonstrated a system for interpreting sign language in order to assist deaf and hearing people, shows that technology can be used to aid their acquisition of sign language [7]. According to the teaching sign language function, the application provides video tutorials about sign language for hearing-impaired kids and allows them to practice English and mathematics using hand gestures. Aligning with the objective, which is identifying letters and numbers. According to this function, get an analysis of students' letter and number writing accuracy.

Another objective of the research is to provide a platform for recall and practice lessons using gamification activities, while students doing the activities on the gamification platform suggest the best way to practice student activities using facial emotion recognition technology. Gamification has gained recognition as an effective approach in educational settings. Another study states how empirical evidence on the impact of gamification in teaching and learning has been systematically reviewed [8]. Their findings indicated positive outcomes, including increased motivation and engagement.

There are several research have for facial recognition technology and teaching in classroom activities. There is research for the Student Emotion Recognition System (SERS) to improve online learning through the measurement of learner concentration [9]. By utilizing facial recognition technology, the application can gauge the child's engagement and attention levels. And another study mentions students' facial expressions during a lecture [10]. Previous studies have highlighted the importance of teaching sign language to facilitate communication and language development. Furthermore, the incorporation of English letters and numbers, learning activity games, and facial recognition techniques can significantly enhance the learning experiences of hearing-impaired children. However, further research and development are necessary to implement and evaluate the effectiveness of such an e-learning application in Sri Lanka's context.

Previous research has shown the potential of e-learning applications to support the educational journey of hearing-impaired children. However, there is a research gap in the development of e-learning applications that combine sign language, gamification activities, and facial emotion recognition technology to provide a comprehensive and engaging learning experience for hearing-impaired children in Sri Lanka. Existing e-learning applications for these children typically focus on teaching sign language or delivering educational content through various modalities but lack integration of these elements. This study aims to bridge these gaps by developing an e-learning application specifically designed for hearing-impaired children in Sri Lanka that incorporates all of these components and assesses its effectiveness in enhancing learning outcomes.

## III. Methodology

In this research, an E-Learning system for hearing-impaired primary school students to learn letters and numbers through the use of four key functionalities is presented. They are, 1. Sign Language Recognition, 2. Identify the handwritten letters


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and numbers, 3. Identifying emotions using facial expressions, 4. Gamified learning environment.

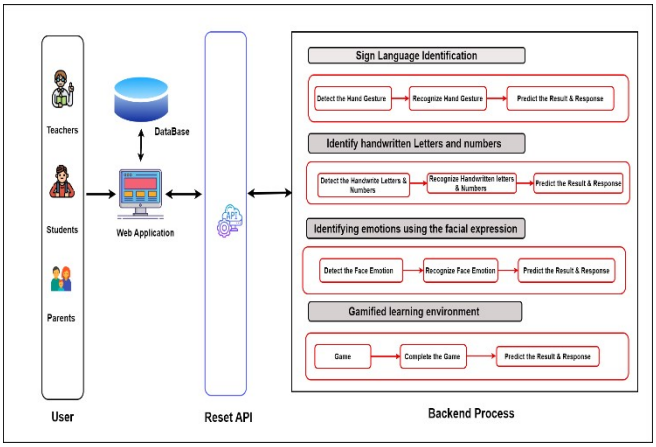



Fig. 1. System overview

### A. Sign Language Recognition

This application allows hearing-impaired students opportunities to learn letters and numbers using sign language, and it can also be a great opportunity for the user to learn and improve their knowledge of sign language. In this application, used the American Sign Language (ASL) alphabet (A-Z) and numbers (0–9). This e-learning platform provides video lessons and interactive images to teach sign language. After being given basic lessons, such as alphabet letters in ASL, the system will give a test to check their knowledge of what they have studied so far. To face the test, a student requires only a personal computer, webcam, and internet connection. After the task begins, the system displays the letter or number of the alphabet on the screen and asks the student to indicate the correct sign language for that letter. The user should do that in front of the webcam. If the user's motion is correct, it is saved, and the user moves on to the next task; however, if the user's motion is incorrect, the student is requested to repeat the task again. "Fig. 2" shows how the system detects the sign of the user.




Fig. 2. Recognize the hand sign

All the images were taken under low-resolution laptop webcams to get the low-resolution images based on training purposes [11]. Furthermore, this model can be used to identify images in real-time with low latency. For successful model training, 750 sign images were collected, and all the images were divided into train and test datasets. Then, for trains and test image datasets, the tested individual images have been converted to CSV (Comma Separated Value) files. Using the CSV files, the model is started to train using the TensorFlow model trainer and Keras library with the CNN (Convolutional Neural Networks) configuration. A convolutional neural network (CNN) is a type of deep learning algorithm that is widely used for image recognition and analysis purposes [15]. After successfully training the model, the webcam opens and detects images using the pre-trained model. Once the image detection is over, with an accuracy 96%, the image will be sent through an API to the image classifier for further classification [11, 12].

In the intricate process of image classification within a Convolutional Neural Network (CNN) classifier, several steps are involved. The first step is pooling, a critical layer in the network's architecture. During pooling, the image is subjected to a series of convolutional layers, each equipped with specialized kernels. These convolutional layers serve to capture important visual patterns and features within the image. Simultaneously, they contribute to the reduction of the image's dimensionality, which is crucial for efficient processing. This process continues with the addition of multiple convolutional layers, with pooling being applied iteratively until the desired level of filtration is achieved. Once this intricate feature extraction is complete, the network then flattens the resulting feature maps into a one-dimensional vector. Subsequently, this vector is passed through a fully connected layer, which employs complex computations to map the features to a specific output class. To determine the class, an activation function is applied, enabling the model to identify the predicted sign letter. Once the sign letter is identified, it is seamlessly transmitted to the Text and Audio Generator section through an API, where it can be converted into the corresponding text or audio representation, facilitating seamless communication for individuals using sign language [11, 14].

### B. Identify the handwritten letters and numbers

Identifying letters and numbers are necessary for this component. There are three categories for both letters and numbers. Numbers and letters are taught to students first. Here, the digits teach the numerals from 1 to 10, and the letters teach the English alphabet from A to Z. Second, they must use sticks to create numbers and letters. At that point, kids will be certain that they grasp the letter or number. Then, students can type letters on the computer screen. Determine whether a digit is right or incorrect after children have written letters.

Use the SVC (Support Vector Clustering) technique to recognize the handwritten text and digits entered by the users. This is constructed by SVM (Support Vector Machine). SVM is useful for a variety of tasks, including regression and classification. Using a kernel function in SVC converts data points from the data space to the high dimensional feature space. Using the support vector domain description technique,

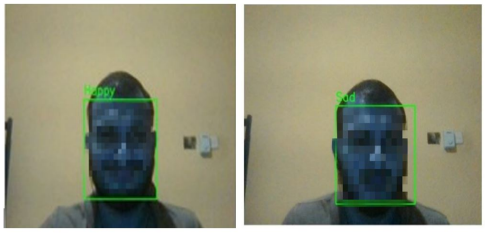

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the kernel's feature space algorithm determines the shortest range that fully encompasses the image of the outcomes. A series of contours that encircle the data points are produced when this range is remapped into the data space. Such contours are interpreted by the SVC as cluster borders, and the points they cover are connected to the same cluster [17].

To identify the numbers, a total of 1000 images were used. To fit the model and determine its accuracy, the system used the Support Vector Clustering algorithm. This approach seeks to arrange data more meaningfully by categorizing it according to specific criteria [16].

### C. Identifying emotions using the facial expression

This study aimed to use facial recognition technology to select the best learning method for students with hearing loss. The participants of the study were 20 students with hearing loss, aged between 5 and 10 years. The participants were randomly assigned to one of three groups, each group playing a different educational game. The games were engaging activities, and students were able to visually identify pictures and select the appropriate word for each image. They will also be able to practice and recall math operations using visual aids, such as graphics and pictures, to enhance their understanding of math operations. The front camera on the device will be automatically accessible by the app when a student begins to play the game in order to capture the student's face and record their facial expressions for the duration of the activity. The application used trained models, such as facial emotion detection models, for analyzing students' facial expressions and emotions. It will primarily identify five emotions: happy, sad, angry, surprised, and neutral. "Fig. 3" shows how to detect the user's emotions. The application also recorded the duration of each emotion and the number of times the student switched to another game.




(a) Emotion - Happy (b) Emotion - Sad

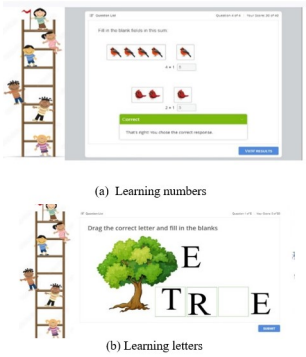
Fig. 3. Sample Emotion Analysis

More than 5,000 photos of kids with various facial expressions between the ages of 5 and 10 were used to construct the dataset for facial emotion identification. The dataset includes a collection of facial expression pictures representing the five basic emotions of happiness, sadness, anger, surprise, and neutrality. The dataset's images were used to train the model. CNN was used to train this model. CNN was chosen for this model's training because it offers a very high level of accuracy in picture categorization and because it does so automatically and without human supervision. When thousands of features need to be retrieved, it is therefore suitable. The sequential model receives the preprocessed images in this research paper. In the model, the fundamental components are the convolutional network, max pool layer, dropout layer, and dense layer networks. The preprocessed photos are fed into the convolution layer and the max pool layer. The convolution layer creates a convolution kernel that is convolved with the layer input, resulting in a tensor of outputs. Following the convolution layers, a max pooling layer with a pool size of 2x2 is added. This layer down-samples the input along its spatial dimensions (height and width) by selecting the maximum value over an input window for each channel of the input [18]. The CNN model has been trained to recognize the five primary emotions of surprise, anger, sadness, neutral and happiness [13].

### D. Gamified learning environment

The component we've developed facilitates a gamified learning environment for hearing-impaired primary students to engage in interactive games. The system utilizing Human-Computer Interaction (HCI) theory, aims to enhance the effective learning of letters and numbers. Given that children may not always be interested in traditional learning, this interactive environment is designed to teach lessons more engagingly. The games within this web-based learning environment feature visually appealing graphics, colors, and images, ensuring an attractive and interactive backdrop for learning. The user interface is designed with simplicity in mind, making it easy for children to navigate and use.

The system is web-based and applies Motivational Design principles, incorporating game elements like points, badges, and leaderboards to motivate students. An image is initially provided in each game, and the student must supply the correct letters corresponding to the picture. In case of an accurate answer, a confirmation message appears, otherwise, an incorrect answer notification is displayed. The game also features number-related activities and a letter-matching exercise, providing a variety of ways for students to practice and reinforce their knowledge.




(a) Learning numbers

(b) Learning letters

Fig. 4. Sample game base learning interfaces (a) Interface for learning numbers (b) Interface for learning letters

The component we've developed employs a User-Centered Design (UCD) approach, taking into account the specific


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needs, preferences, and contexts of hearing-impaired primary students. We've designed this component to provide an educational gamified learning environment, leveraging cutting-edge technologies like React and Python as depicted in "Fig. 4"

## IV. Result & Discussion

This section elaborates on the experiment outcomes for each model. This research aims to introduce a more effective E-learning platform for hearing-impaired primary school children.

In times of crisis, hearing-impaired students need an effective way to continue their education. It would not be effective for those with hearing impairment to use an online teaching model, where teachers speak, and students listen. Therefore, as an option for deaf and hard-of-hearing students, the e-learning platform has been proposed to support them in learning letters and numbers with proper instructions Python is used for the application's backend, and the front end of the application is built using React, an SQLite database, Flask, and TensorFlow. Machine learning concepts and algorithms, image processing, face recognition techniques and gamification environments have been used in the research.

### A. Sign Language Recognition

CNN is used for image processing and recognition. The alphabet and the numbers [1–10] were the two categories that were selected for training. To detect the alphabet, the CNN model was trained with 26 classes where each class has 750 images. From the data set, 80% of the samples were used for training and 20% of the samples were used for testing purposes. For the test input data, the CNN model achieved an overall 96% accuracy as shown in "Fig. 5" and 0.1251 test error in the training process with 50 epochs of the model. "Fig. 6" presents the Precision, Recall, F1 score, and accuracy for each letter in the trained model's classification.

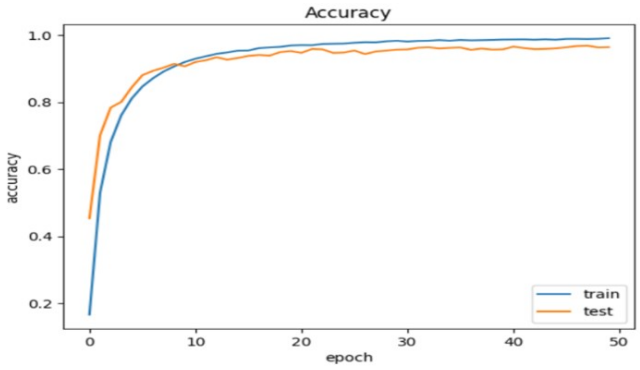



Fig. 5. Training and Validation Accuracy graph of letter Recognition

Using the MNIST number dataset, CNN was trained to recognize the numbers with an accuracy of 97%. For improved training, the epochs were set to 50, and "Fig. 7" displays the training and validation accuracy.

| Letter | Precision | Recall | F1 Score |
|---|---|---|---|
| A | 1 | 1 | 1 |
| B | 1 | 1 | 1 |
| C | 1 | 1 | 1 |
| D | 0.998331 | 1 | 0.999165 |
| E | 1 | 0.998955 | 0.999477 |
| F | 1 | 1 | 1 |
| G | 1 | 0.990826 | 0.995392 |
| H | 0.990225 | 1 | 0.995088 |
| I | 1 | 1 | 1 |
| K | 1 | 1 | 1 |
| L | 1 | 1 | 1 |
| M | 1 | 0.998104 | 0.999051 |
| N | 1 | 1 | 1 |
| O | 1 | 1 | 1 |
| P | 1 | 1 | 1 |
| Q | 1 | 1 | 1 |
| R | 0.996909 | 0.996909 | 0.996909 |
| S | 0.997504 | 1 | 0.998751 |
| T | 1 | 1 | 1 |
| U | 0.998279 | 0.999139 | 0.998709 |
| V | 1 | 0.997227 | 0.998612 |
| W | 1 | 1 | 1 |
| X | 1 | 1 | 1 |
| Y | 1 | 1 | 1 |

Fig. 6. Model accuracy

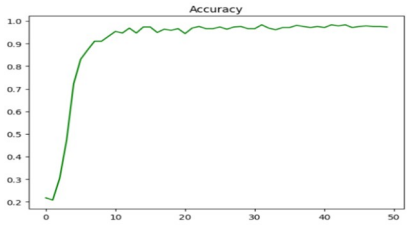



Fig. 7. Training and Validation Accuracy graph of Numbers Recognition

### B. Identify the handwritten letters and numbers

More than 1,000 letters and numbers were collected for each letter and number for recognition. Machine learning is used to identify the numbers and letters that children write. This dataset was split into two sets 20% was utilized for testing and 80% for training. The Support Vector Clustering algorithm has been applied to the model training process to fit it and verify its accuracy. For the test input data, the SVM model achieved above 98% accuracy.

### C. Identifying emotions using the facial expression

The dataset collection consisted of more than 5,000 facial expression images of students, which were used to train a CNN model with a resulting accuracy of 98% and a loss of 2%, as depicted in "Fig. 8" The accuracy can be improved by feeding more training data to the neural network. The model is capable of detecting five common emotions among students, including surprise, anger, sadness, neutral and happiness.

### D. Gamified learning environment

According to the gamified learning environment, when a child completes a level successfully, he is unlocked to the next level and the child gets a reward according to the points. "Fig. 9" shows the result of the game.


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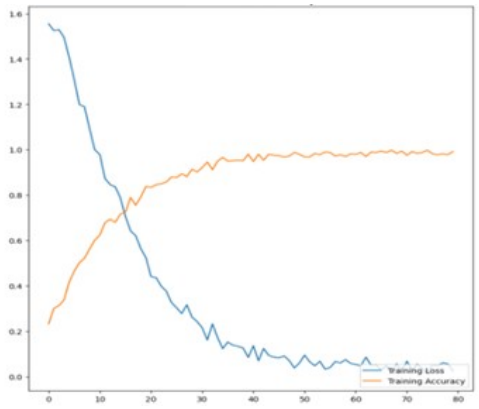



Fig. 8. Loss and Validation Accuracy graph of Face emotion detection

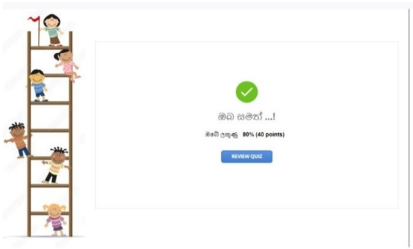


Fig. 9. Results of the game

## V. CONCLUSION AND FUTURE WORK

In Sri Lanka, there is a shortage of E-learning applications developed for hearing-impaired students to learn sign language, basic letters, and numbers more easily. The study's output was the creation of the "Visual Kids" e-learning platform, which offers primary school-aged children with hearing impairments in Sri Lanka a successful learning environment for learning letters and numbers.

The system was built using technologies like React, SQLite, Flask, TensorFlow, and Python, incorporating machine learning and image processing techniques. The evaluation of the sign language recognition model showed 96% accuracy in detecting alphabet signs and 97% accuracy for number signs. The handwriting recognition model achieved 98% accuracy. The facial emotion recognition model can detect five emotions with 98% accuracy.

By leveraging technology to create an accessible, interactive, and adaptable environment, this e-learning platform shows the potential to significantly enhance educational outcomes for hearing-impaired students. It can facilitate independent learning and provide engagement analytics to teachers.

However, more extensive testing and feedback from students and teachers should be conducted to further refine the system. Additional features could also be incorporated, such as expanding the curriculum beyond letters and numbers, adding more sign languages, and extending it to higher grade levels. Overall, this study provides a critical proof of concept and a starting point for creating technologically-enabled strategies to support inclusive education for Sri Lankans with hearing disabilities.

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